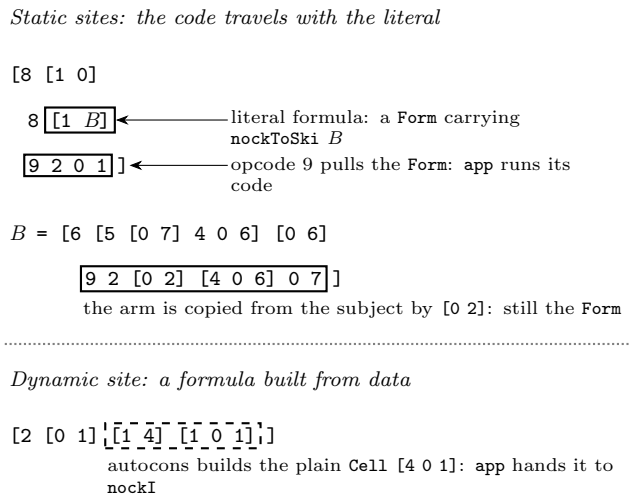
`nockmark`: annotated code fragments

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v0.1.0, 2026-11-06

`nockmark` draws a code fragment as a column of lines with boxed spans and notes, using TikZ. It was made for Nock, which has no layout of its own, but any monospaced code works. It runs under pdfLaTeX, XeLaTeX and LuaLaTeX.

1 Example



The source:

```
\begin{nockmark}
  \nkheading{Static sites: the code travels with the literal}
  \nkline{\nk{[8 [1 0]}}
  \nkline[1]{\nk{8 }\nkstatic{s1}{[1 \nkmeta{B}]}}
  \nkline[1]{\nkstatic{s2}{9 2 0 1}\nk{]}}
  \nknote{s1}{literal formula: a \texttt{Form} carrying ...}
  \nknote{s2}{opcode~9 pulls the \texttt{Form}: ...}
  \nkskip{0.3}
  \nkline{\nk{\nkmeta{B} = [6 [5 [0 7] 4 0 6] [0 6]}}
  \nkline[4]{\nkstatic{s3}{9 2 [0 2] [4 0 6] 0 7}\nk{]}}
  \nknotebelow{s3}{the arm is copied from the subject ...}
  \nkrule
  \nkheading{Dynamic site: a formula built from data}
  \nkline{\nk{[2 [0 1] }\nkdynamic{d1}{[1 4] [1 0 1]}\nk{]}}
  \nknotebelow{d1}{autocons builds the plain \texttt{Cell} ...}
\end{nockmark}
```

2 Commands

`\begin{nockmark}[keys]` a fragment; it is a `tikzpicture`, so place it in a `figure` or `center`.

`\nkline[`*n*`]{`*pieces*`}` a line at the next baseline, indented n steps.

`\nk{`*code*`}` plain code. A trailing space is kept.

`\nkstatic{`*name*`}{`*code*`}` a solid box, named for notes.

`\nkdynamic{`*name*`}{`*code*`}` a dashed box.

`\nkbox[`*style*`]{`*name*`}{`*code*`}` a box in any TikZ style, e.g. one defined with `\tikzset`.

`\nkmeta{`*x*`}` a metavariable inside code, in the text face.

`\nknote[`*opts*`]{`*name*`}{`*text*`}` a note in the right column, with an arrow to the end of the box's line.

`\nknotebelow[`*opts*`]{`*name*`}{`*text*`}` a note under the box; following lines clear it.

`\nkheading{`*text*`}`, `\nkrule`, `\nkskip{`*cm*`}` a heading, a dotted divider, extra space.

`\nockmarkset{`*keys*`}` set keys for every later fragment.

3 Keys

Lengths without a unit are in centimetres. Defaults in parentheses.

`line sep` (0.6) baseline to baseline.

`indent` (0.25) one indent step.

`heading sep` (0.75) heading baseline to the next line.

`note column` (2.9) left edge of right-column notes.

`note width` (4.9cm), `below width` (7.2cm) note widths.

`below skip` (0.45) from a below-note's bottom to the next baseline.

`rule raise` (0.15), `rule skip` (0.4) rule placement.

`box gap` (1.6pt) box padding and spacing.

`code font`, `note font`, `heading font`, `meta font` the faces.

The TikZ styles `nk code`, `nk box`, `nk static`, `nk dynamic`, `nk note`, `nk heading`, `nk arrow` and `nk rule` can be redefined with `\tikzset`.

4 Other code

```
++  dec

  |=  a=@

  ?<  =(0 a)  !!
  =+  b=0

  |-  ?:  =(a +(b))  b
  $(b +(b))
```

crash on zero

the loop test